MILITARY SPECIFICATION

HEELS, ORTHOPEDIC, CAST

This specification is approved for use by all Departments and Agencies of Department of Defense.

1. SCOPE

1.1 Scope. This specification covers adult size walking heels for incorporation into an orthopedic cast of the lower leg.

1.2 Classification. The orthopedic cast heels covered by this specification shall be of the following types, as specified (see 6.3):

Type I. - Firm walking surface; triangle design tread.
Type II. - Medium firm walking surface; fine serrated tread.
Type III. - Firm walking surface; course serrated tread.
Type IV. - Resilient walking surface; diamond design tread.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-566 - Box, Folding, Paperboard.
PPP-B-585 - Box, Wood, Wirebound.
PPP-B-601 - Box, Wood, Cleated-Plywood.
PPP-B-621 - Box, Wood, Nailed and Lock-Corner.
PPP-B-636 - Box, Shipping Fiberboard.
PPP-B-676 - Box, Setup.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Hq, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 2800 S. 20th St., Phila., PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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MIL-P-116 - Preservation, Methods Of.

MIL-L-10547 - Liners, Case, and Sheet, Overwrap; Water-vaporproof or Waterproof, Flexible.

STANDARDS

FEDERAL

Fed. Test Method Std. No. 601 Rubber, Sampling and Testing.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

DRAWINGS

Defense Personnel Support Center (DPSC)

No. 23181 HEEL, ORTHOPEDIC CAST: ZIMMER DESIGN (Type II).
No. 23182 HEEL, ORTHOPEDIC CAST, STRYKER DESIGN (Type III).
No. 23183 HEEL, ORTHOPEDIC CAST, WRIGHT DESIGN (Type IV).
No. 23184 HEEL, ORTHOPEDIC CAST, ORTHOPEDIC EQUIPMENT DESIGN (Type I).

(Copies of specification, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials.

3.1.1 Rubber. Types I, II and IV shall be molded from either natural or synthetic rubber compound suitable in all respects for the purpose intended. Properties shall be as indicated in table I.

3.1.2 Plastic. Type III shall be molded from ethylene propylene as indicated in table I.

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TABLE I. Physical Properties

Type	Material	Durometer hardness (shore A)	Tensile strength (min. psi)	Elongation minimum
I	Rubber	65-75	1000	350%
II	Rubber	55-65	1700	400%
III	Plastic	65-75	1550	700%
IV	Rubber	55-65	1500	350%

3.2 Construction. Style and design shall be in conformance with the drawing for the applicable type as specified (see 2.1). The heel shall be an all-rubber, or all-plastic when specified (see 2.1), walking heel consisting essentially of a flat plate and heel block centered thereon. Heel shall be molded of one piece construction. The walking surface of the heel shall be curved to provide "rocker" action and treaded to prevent skidding when in use. The heel shall be so constructed that it may be easily incorporated in the weight-bearing surface of a walking plaster cast of the lower leg. The rubber (or plastic) portion which anchors heel to the cast shall be sufficiently rigid for the intended purpose.

3.3 Color. Color shall be manufacturers' commercial color except black and other floor marking colors shall not be used.

3.4 Identification marking. The manufacturer's name or registered trademark shall be clearly and legibly molded into the heel.

3.5 Workmanship. Workmanship shall be first class throughout. Heel shall be free from defects which detract from its appearance or may impair its serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

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4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the rubber (or plastic), may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer of supplier, the National Stock Number (NSN), the item identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and as indicated in table II. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics. The unit of product for examination purposes shall be one heel.

TABLE II. Sampling criteria for tests.

Type of defect	Inspection level	AQL (Percent defective)
Major	II	2.5
Minor	II	6.5
Dimensional	S-3	4.0

4.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, and as indicated in Table III. Unit of product for test purposes shall be one heel.

TABLE III. Sampling criteria for tests.

Component	Characteristic	Requirement	Test procedure	Insp. level	AQL
Rubber (Plastic)	Hardness	Table I	Table V	S-2	1.0
	Tensile strength	Table I	Table V	1/	2/
	Elongation	Table I	Table V	1/	2/

1/ Two sheets, 6 x 6 x .060 inches from each lot of rubber or plastic used in the manufacture of the heels.
2/Acceptance number shall be zero.

4.2.1 Examination. The heels shall be examined for defects including, but not limited to, those listed in table IV. Classification of additional defects shall be in accordance with MIL-STD-105. Any heel sample containing one or more defects shall be rejected and if the number of defectives in any sample exceeds the acceptance number for the sample the lot represented by that sample shall be rejected.

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TABLE IV. Classification of defects.

Categories	Defects
Major A	
101	Heel not free of cracks and splits.
102	Incorrect tread design.
103	Identification marking omitted, incomplete or illegible.
104	Tread design not clearly and sharply defined.
Minor	
201	Grease, oil stains and foreign matter not readily removed.
202	Excess flash not removed.
203	Edges not sharply defined

4.2.1.1 Dimensional examination. Any dimension that is not within the tolerance specified shall be classified as a defect.

4.2.2 Packaging inspection. The inspection of the preservation-packaging and interior package marking shall be in accordance with the group A quality conformance inspection requirements of MIL-P-116. The inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirement of MIL-STD-129.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements. Test methods shall be as indicated in table V.

TABLE V. Test Methods.

Test	Method
Tensile strength	11011 of Fed. Test Method Std. 601
Elongation	11021 of Fed. Test Method Std. 601
Durometer hardness	3021 of Fed. Test Method Std. 601

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.

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5.1.1.1 Cleaning. Each heel shall be cleaned in accordance with MIL-P-116, process C-1.

5.1.1.2 Unit packaging. Each heel shall be packaged in a box of appropriate size constructed in accordance with PPP-B-566 or PPP-B-676. Closure of box shall be as specified in the appendix of the applicable box specification. Packaging shall be method III in accordance with MIL-P-116.

5.1.1.3 Intermediate packaging. Twelve (12) heels shall be packaged in a box of appropriate size and design constructed in accordance with PPP-B-566, PPP-B-676 or PPP-B-636, type CF, class domestic. Closure shall be accordance with the appendix of the applicable box specification.

5.1.2 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. One hundred forty four (144) heels shall be packed in a box designed for a type 2 load, and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, Class 2 or PPP-B-636, class weather resistant, grade V3c. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Fiberboard box shall be waterproof in accordance with the appendix of PPP-B-636.

5.2.2 Level B. One hundred forty four (144) heels shall be packed in a box designed for a type 2 load, and constructed in accordance with PPP-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, type CF, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Unit, intermediate and exterior container. Each unit, intermediate and exterior container shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. The heel is intended for use as an aid in ambulation and weight bearing after incorporation into the orthopedic cast of the lower leg. The various types are representative of different designs and degrees of tread firmness.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Type required
c. Applicable levels of packaging and packing.
d. National Stock Number.

6.3 This specification is intended to cover the following items listed in the Federal Supply Catalog:

National Stock Number	Item Identification	Type
6515-00-106-8475	HEEL, ORTHOPEDIC CAST, Rubber, Orthopedic Equipment Design, Adult Size	I
6515-00-106-8476	HEEL, ORTHOPEDIC CAST, Rubber, Zimmer Design, Adult Size	II
6515-00-106-8477	HEEL, ORTHOPEDIC CAST, Plastic, Stryker Design, Adult Size	III
6515-00-106-8478	HEEL, ORTHOPEDIC CAST, Rubber, Wright Design, Adult Size.	IV

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6515-2030

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R399

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-H-37577 HEED, ORTHOPEDIC CAST

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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